AI Photo Mentor: A Mobile Vision-Language System for Automated Photography Critique and Skill Development

***Abstract*—Expert photography critique is scarce and costly, limiting skill development for aspiring photographers. We present AI Photo Mentor, a mobile application delivering automated, professional-grade critiques via a fine-tuned Qwen2-VL-2B Vision-Language Model. The system integrates RPCD-Elite-3504, a curated 3,504-sample critique dataset; LoRA fine-tuning with 4-bit quantization; and a native Android application providing real-time capture, critique display, and progress tracking. The model achieves 98.4% JSON validity and 0.72 Pearson correlation with expert evaluators. A repeated-measures user study ($n = 9$) confirmed statistically significant improvement in photographic quality after a single AI feedback cycle ($t(8) = 3.36$, $p = 0.010$, $\Delta = +0.74$), with composition showing the largest gain ($\Delta = +1.11$, $p = 0.021$). Deployed on CPU-only infrastructure via GGUF quantization, critiques are delivered in under 15 seconds at under $0.01 per image.**

***Index Terms*—Vision-Language Models, Mobile Computing, Photography Education, Deep Learning, Android, LoRA, Quantization, User Study**

I. Introduction

Photography has emerged as one of the most accessible forms of visual communication in the digital age. With billions of smartphones equipped with high-quality cameras, photography has been democratized at an unprecedented scale. However, access to quality photography education and constructive feedback remains limited, particularly for enthusiasts in remote areas or those with constrained financial resources.

Traditional photography education relies on expert critique [2]—detailed feedback from experienced photographers analyzing composition, lighting, technical execution, and artistic merit. While essential for skill development, such critique is typically available only through expensive courses, workshops, or mentorship programs that many learners cannot access. This access gap represents a significant barrier to photography skill development.

Recent advances in Vision-Language Models (VLMs) [3] present opportunities to democratize photography education. State-of-the-art systems including GPT-4V, Claude 3, and open-source alternatives such as Qwen2-VL have demonstrated strong capabilities across visual understanding and language generation tasks [3]. However, deploying these models for practical educational applications faces three principal challenges: (1) the scarcity of structured datasets for training domain-specific critique models; (2) computational requirements that exceed the capabilities of mobile hardware; and (3) the need for reliable, structured output generation suitable for educational interfaces.

This paper presents AI Photo Mentor, a complete system addressing these challenges through: (1) curation of the RPCD-Elite-3504 dataset from the RPCD source corpus [1]; (2) parameter-efficient fine-tuning of Qwen2-VL-2B with 4-bit quantization; (3) CPU-efficient deployment via GGUF format and llama.cpp; and (4) a native Android application providing camera capture, critique display, and learning management. Critically, we evaluate the system not only through technical benchmarks but through a controlled user study measuring whether AI critique actually improves photographic output.

Our contributions are as follows:

1) **RPCD-Elite-3504**: the first publicly available structured multimodal dataset for photography critique.
2) **Efficient fine-tuning**: a demonstration that LoRA fine-tuning achieves domain-specific expertise at minimal computational cost.
3) **CPU-efficient deployment**: validation that quantized VLMs maintain acceptable quality on CPU-only infrastructure.
4) **Mobile architecture**: a reference Clean Architecture design for mobile-AI integration.
5) **Empirical user study**: a repeated-measures experiment with nine participants confirming statistically significant improvements in photographic quality after a single round of AI feedback.

II. Related Work

A. Photography Analysis Systems

Commercial photography tools provide limited AI assistance. Adobe Lightroom's Sensei AI [11] offers technical corrections such as exposure and white balance adjustment, but lacks the natural language educational feedback necessary for skill development. EyeEm Vision [12] rates aesthetic quality without producing detailed critique or actionable guidance. Academic systems such as PhotoCritique [10] generate textual feedback but rely on limited visual attribute detection rather than the holistic semantic understanding afforded by modern

VLMs. None of these systems has been evaluated through user studies measuring actual skill improvement.

B. Vision-Language Models

VLMs combine visual perception modules with generative language components, enabling models to reason jointly over images and text. LLaVA [4] pioneered instruction tuning for multimodal dialogue. InternVL [5] scaled vision foundation models to support diverse vision-language tasks. Qwen2-VL [3] introduces native high-resolution processing at 448×448 pixels, efficient visual token representation using 256 tokens per image, and an extended context window. This work builds on the Qwen2-VL-2B-Instruct variant, which balances capability with practical deployability. While larger models (7B–72B parameters) offer superior reasoning, the 2B variant is necessary to meet real-world deployment constraints on consumer-grade infrastructure.

C. Parameter-Efficient Fine-Tuning

Standard fine-tuning modifies every weight in a model, making it computationally prohibitive for large-scale architectures. LoRA [6] inserts compact, trainable decomposition matrices alongside frozen pretrained weights, dramatically reducing the number of updated parameters without full retraining. QLoRA [7] combines LoRA with aggressive weight quantization, making it practical to adapt very large models on commodity GPU hardware. Recent surveys [8] confirm LoRA's effectiveness for multimodal domain adaptation.

D. Mobile AI Deployment

On-device inference [15] leverages mobile neural processing units for lightweight models but cannot accommodate billion-parameter VLMs within current hardware constraints. Cloud-based inference offloads computation but introduces latency and connectivity dependencies. Hybrid approaches [9] combine edge preprocessing with cloud inference to balance responsiveness and capability. AI Photo Mentor adopts cloud-based deployment with CPU-optimized quantization (GGUF) to achieve cost-effective operation without requiring dedicated GPU servers.

E. Research Gap

Existing work either lacks structured critique output, requires prohibitive GPU infrastructure, or fails to address complete mobile integration. Crucially, no prior work provides both an end-to-end pipeline from dataset curation through mobile deployment *and* empirical evidence that AI-generated feedback produces measurable improvement in user photographic output. This paper addresses that gap.

III. Methodology

A. System Architecture

AI Photo Mentor comprises three integrated subsystems:

Mobile Client: A native Android application built with Kotlin and Jetpack Compose, providing CameraX-based real-time preview, critique display, local persistence via Room, and network communication via Retrofit. The application implements Clean Architecture, with strict separation between the Presentation layer (UI and ViewModels), Domain layer (models and repository interfaces), and Data layer (database, network, and repository implementations).

Cloud Inference Service: A FastAPI-based REST service hosting the fine-tuned Qwen2-VL model. The service accepts image uploads, performs VLM inference via llama.cpp, and returns structured JSON critiques. Model weights use GGUF Q4_K_M quantization to enable CPU-efficient inference on entry-level cloud instances.

Data Management: A Room database manages critique history and progress tracking on the device. Cloud storage hosts model artifacts and dataset resources.

B. Dataset Development

The RPCD-Elite-3504 dataset was curated from the Reddit Photo Critique Dataset (RPCD) [1], which comprises 31,983 posts from r/photocritique spanning 2012–2024.

Filtering Criteria: Posts were retained only if they had a minimum of three community comments, a Reddit score of at least 10, a successfully downloaded image, and passed deduplication via perceptual hashing. After filtering, the top 5,000 highest-scoring posts were selected as candidates. Of these, 3,504 images were successfully retrieved and processed through the multimodal annotation pipeline; the remainder were lost to dead image links, download failures, or duplicate detection, yielding the final RPCD-Elite-3504 corpus. Genre distribution in the final corpus was approximately: Portrait (25%), Landscape (20%), Street (15%), Wildlife (12%), Architecture (10%), Macro (8%), Event (6%), and Abstract (4%).

Structured Annotation: Gemma 4 31B Instruct was used to extract structured critiques from the Reddit comment threads, producing: (1) a composition score (1–10) and written critique; (2) a lighting score (1–10) and written critique; (3) a technical score (1–10) and written critique; (4) an overall score (1–10); and (5) three to five actionable improvement steps. Human validation confirmed annotation quality with a Cohen's κ of 0.72.

TABLE I
RPCD-Elite-3504 Dataset Statistics

Metric	Value
Total Samples	3,504
Training Set	3,153 (90%)
Validation Set	176 (5%)
Test Set	175 (5%)
Average Resolution	2048×1536
Mean Overall Score	6.90 ($\sigma = 1.23$)

The overall score distribution is right-skewed toward quality, with a mean of 6.90 ($\sigma = 1.23$), reflecting the post-filtering bias toward highly-upvoted community submissions. Scores range from 2 to 10, with 68% of samples concentrated between scores 6 and 8.

C. Model Fine-Tuning

We fine-tuned Qwen2-VL-2B-Instruct using the Unsloth library [13] on Kaggle's dual NVIDIA T4 GPU environment (16 GB VRAM each).

Data Preparation: Training samples were formatted as ChatML conversations with a system prompt ("You are a professional photography critic"), a user prompt consisting of the image and the instruction "Analyze this photograph", and an assistant response containing a structured JSON critique.

Quantization: Using bitsandbytes, the base model weights were compressed to 4-bit precision under an fp16 compute regime, cutting peak GPU memory consumption from roughly 8 GB to under 2 GB.

LoRA Configuration: Low-rank adapters with rank $r = 16$ ($\alpha = 16$) were applied to all major components: the vision encoder, the language decoder, attention projection matrices, and MLP layers. This comprehensive adaptation enabled effective learning across both visual feature extraction and critique generation, while requiring only a fraction of the parameters updated by full fine-tuning.

TABLE II
FINE-TUNING HYPERPARAMETERS

Parameter	Value
Learning Rate	2×10^{-4}
Batch Size (effective)	16
Epochs	3
LoRA Rank (r)	16
LoRA Alpha (α)	16
Dropout	0
Warmup Steps	50
LR Scheduler	Cosine
Optimizer	AdamW (8-bit)

D. Deployment Optimization

After training completed, the learned low-rank updates were absorbed back into the original weight matrices. The complete model was exported to GGUF format [14] as two separate files required by llama-cpp-python: (1) the language model quantized to Q4_K_M (~1.3 GB) for fast CPU decoding, and (2) the multimodal projector (mmproj) and vision encoder retained in F16 (~400 MB). Together these total under 2 GB, enabling CPU-only inference with less than 5% degradation in expert correlation compared to the full-precision model.

E. Mobile Application

The Android application implements six functional modules:

Camera: CameraX provides a real-time preview with configurable grid overlay, flash control, and Photo Picker integration. Images are resized to a maximum dimension of 1024 px before upload, reducing payload size by approximately 60%.

Critique Display: Animated score rings present the overall score and each sub-dimension score, complemented by expandable critique cards and actionable improvement checklists. The interface uses a dark-themed glassmorphism design for clarity.

Gallery: Room database persistence supports offline access to critique history, presented as a chronological thumbnail gallery with date filtering.

Learning: Eight curated lessons cover composition (Rule of Thirds, Leading Lines, Framing, Symmetry), lighting (Golden Hour, Portrait Lighting), technical skills (Exposure Triangle), and creative techniques (Minimalism).

Progress: Statistical aggregation surfaces daily, weekly, and all-time metrics including average scores, best scores, category trends, and usage streaks to encourage sustained engagement.

Settings: Dynamic endpoint configuration, grid toggle, and lesson-completion tracking allow customization of the user experience.

IV. EVALUATION

A. Technical Benchmark

Fine-tuning completed in 45 minutes over three epochs. Training loss decreased monotonically from 2.5 to 0.8, with no signs of overfitting. Table III summarizes model performance on the held-out test set ($n = 175$).

TABLE III
MODEL EVALUATION RESULTS ON TEST SET ($n = 175$)

Metric	Value
JSON Validity	98.4%
Score Range Adherence	100%
Average Response Length	380 tokens
Expert Correlation (Pearson r)	0.72
Mean Absolute Error (Overall Score)	0.82 points
± 1 Score Agreement	78%

Per-Dimension Correlations: Pearson correlation with expert scores was: Overall $r = 0.74$, Technical $r = 0.73$, Composition $r = 0.71$, Lighting $r = 0.69$. All correlations were statistically significant ($p < 0.001$).

Expert Review: Five experienced photographers independently rated a sample of 50 AI-generated critiques on a 1–5 scale. Mean ratings were: Critique Accuracy 4.2/5.0, Actionability 4.0/5.0, Technical Correctness 4.3/5.0, and Tone 4.5/5.0.

B. Inference Performance

Table IV reports end-to-end inference latency across hardware configurations. CPU deployment achieves approximately four images per minute, which is adequate for individual educational use. The mobile application reduces perceived latency through a three-step progress animation (Resizing → Uploading → Analyzing). The inference service is implemented as a FastAPI REST endpoint, with the public URL dynamically distributed to the Android client via a Hugging Face dataset repository, enabling zero-configuration server updates.

TABLE IV
INFERENCE LATENCY BY HARDWARE CONFIGURATION

Hardware	First Token	Total Time
T4 GPU (full precision)	$\sim$2 s	$\sim$8 s
CPU (GGUF Q4_K_M)	$\sim$4 s	$\sim$15 s
M1 Mac (Metal)	$\sim$3 s	$\sim$10 s

C. Deployment Cost Analysis

CPU-based inference on entry-level cloud instances ($20–50 per month) supports over 5,000 critiques per month at approximately $0.01 per critique. This compares favorably to commercial vision API pricing of $0.01–0.03 per image, and enables sustainable operation for educational institutions and non-profit deployments.

D. Comparison with Baselines

Compared to the zero-shot Qwen2-VL-2B baseline, structured output validity improved from 65% to 98.4%, and expert correlation improved from 0.45 to 0.72. Compared to GPT-4V, the fine-tuned model achieves comparable output validity (98.4% vs. 95%) at approximately 90% lower operational cost.

V. USER STUDY

A. Study Design

To assess whether AI-generated critique translates into measurable improvement in photographic output, we conducted a controlled repeated-measures user study. Nine participants (Sessions 1–9) were recruited from the university community. No formal photography training was required or excluded. Each participant selected a subject of their own choosing; subjects varied across participants.

Each participant completed a single session structured as follows:

1) **Pre-feedback photograph (Photo ID x.1):** The participant freely chose a subject of their own choosing and captured an initial photograph using the AI Photo Mentor application's camera module.
2) **AI critique review:** The application returned structured feedback, including composition, lighting, and technical scores, a written critique for each dimension, and a list of specific improvement suggestions.
3) **Post-feedback photograph (Photo ID x.2):** The participant re-photographed their own chosen subject, aiming to apply the AI-generated suggestions received in step 2.

Photographs were scored on composition, lighting, and technical execution, each on a scale of 1–10. An overall score was calculated as the mean of the three dimensions. Scores were assigned by the AI system, consistent with its use as the evaluation instrument throughout the study.

TABLE V
PER-SESSION SCORES BEFORE AND AFTER AI FEEDBACK

Session	Photo	Comp.	Light.	Tech.	Overall
1	1.1	4	5	5	4.67
	1.2	7	6	7	6.67
2	2.1	5	7	6	6.00
	2.2	7	8	7	7.33
3	3.1	5	7	6	6.00
	3.2	5	7	6	6.00
4	4.1	7	7	7	7.00
	4.2	7	8	7	7.33
5	5.1	5	7	6	6.00
	5.2	6	7	6	6.33
6	6.1	5	7	6	6.00
	6.2	7	6	7	6.67
7	7.1	7	8	6	7.00
	7.2	7	8	7	7.33
8	8.1	7	8	6	7.00
	8.2	7	8	7	7.33
9	9.1	5	6	6	5.67
	9.2	7	8	6	7.00

B. Results

Table V reports all individual scores before and after feedback. Table VI presents aggregate statistics and paired t-test results.

Overall Quality. Mean overall score increased from 6.15 ($\sigma = 0.77$) to 6.89 ($\sigma = 0.50$), a gain of 0.74 points ($t(8) = 3.36$, $p = 0.010$). Eight of nine participants recorded an improvement or no change; one participant (Session 3) showed no change across any dimension.

Composition. The largest and most consistent improvement was observed in composition, which improved by a mean of 1.11 points ($t(8) = 2.86$, $p = 0.021$). This finding suggests that composition guidance—such as rule-of-thirds framing and subject placement—is among the most immediately actionable feedback the system provides.

Technical Execution. Technical scores improved by a mean of 0.67 points ($t(8) = 2.83$, $p = 0.022$), indicating that exposure, focus, and sharpness-related guidance was also effectively communicated and applied.

Lighting. The improvement in lighting scores (mean $\Delta = +0.44$) did not reach statistical significance ($p = 0.169$). This is consistent with the expectation that lighting is harder to adjust in a single session, as it often depends on environmental conditions beyond a novice photographer's immediate control.

Largest Individual Gains. Session 1 showed the largest improvement (+2.00 overall), starting from the lowest baseline score (4.67) and reaching 6.67 after feedback. Session 2 and Session 9 each improved by 1.33 points overall.

C. Discussion

The user study results support two main conclusions. First, AI Photo Mentor's critique is actionable: participants were

TABLE VI
SUMMARY STATISTICS AND PAIRED t-TEST RESULTS ($n = 9$ SESSIONS)

Dimension	Mean Before	SD Before	Mean After	SD After	Mean Δ	p-value
Composition	5.56	1.13	6.67	0.71	+1.11	0.021*
Lighting	6.89	0.93	7.33	0.87	+0.44	0.169
Technical	6.00	0.50	6.67	0.50	+0.67	0.022*
Overall Score	6.15	0.77	6.89	0.50	+0.74	0.010*

* Statistically significant at $\alpha = 0.05$ (two-tailed paired t-test).

able to identify and apply compositional and technical improvements within a single feedback cycle. Second, gains are most pronounced for participants who begin from a lower skill baseline (Sessions 1, 2, 9), consistent with the well-established finding in education research that novice learners benefit most from structured, specific feedback [16].

The absence of significant improvement in lighting is not indicative of a system failure. Lighting is an environmental variable that may require repeated sessions, dedicated practice, or changed shooting conditions to improve measurably. Future longitudinal studies should examine whether lighting scores improve over multiple sessions.

The primary constraint of this study is its limited participant pool ($n = 9$), which reduces statistical power and restricts the scope of generalisation. A full-scale evaluation with a larger and more demographically diverse participant pool, a control group receiving no feedback, and multiple sessions per participant would more rigorously establish causal claims about learning efficacy. We present these results as a promising pilot indicating that further study is warranted.

VI. Conclusion

This paper presented AI Photo Mentor, a complete pipeline for automated photography critique using fine-tuned Vision-Language Models and mobile deployment. We made five principal contributions: the RPCD-Elite-3504 dataset, a demonstration of cost-effective LoRA fine-tuning for visual critique, CPU-efficient GGUF deployment, a Clean Architecture mobile reference implementation, and a pilot user study confirming statistically significant improvements in photographic quality after a single round of AI feedback.

Limitations. The dataset reflects the demographic composition of Reddit's photography community. Larger models may produce better critique on ambiguous or stylistically unusual images. The system currently requires network connectivity for inference. The user study sample is small and lacks a control group.

Future Work. Planned extensions include: expanding the dataset to improve demographic and stylistic diversity; on-device inference via model distillation; integration with formal photography curricula; and longitudinal studies tracking skill development over multiple sessions and weeks. A randomized controlled trial comparing AI feedback to no feedback and to human expert feedback would provide the strongest evidence for educational effectiveness.

The results presented here demonstrate that professional-quality AI critique can be delivered cost-effectively at scale, contributing to the broader goal of democratizing photography education.

Acknowledgment

The authors acknowledge the nine volunteers who participated in the user evaluation are warmly thanked for their involvement. The authors also recognise the open-source community behind Qwen2-VL, Unsloth, and llama.cpp.

References

[1] D. V. Nieto, L. Celona, and C. Fernandez-Labrador, "Understanding Aesthetics with Language: A Photo Critique Dataset for Aesthetic Assessment," in *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, vol. 35, 2022.

[2] T. Barrett, *Criticizing Photographs: An Introduction to Understanding Images*. McGraw-Hill, 2000.

[3] P. Wang et al., "Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution," *arXiv:2409.12191*, 2024.

[4] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual Instruction Tuning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[5] Z. Chen et al., "InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 24185–24198, 2024.

[6] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," in *Proc. International Conference on Learning Representations (ICLR)*, 2022.

[7] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[8] Z. Han et al., "Parameter-Efficient Fine-Tuning for Large Models: A Comprehensive Survey," *arXiv:2403.14608*, 2024.

[9] Y. Xu et al., "Efficient Mobile Vision-Language Models via Edge Compression," *IEEE Trans. Mobile Comput.*, 2023.

[10] L. Zhang et al., "PhotoCritique: Automated Photography Feedback Generation," in *Proc. ACM CSCW*, 2019.

[11] Adobe, "Adobe Lightroom Sensei AI Documentation," 2024.

[12] EyeEm, "EyeEm Vision: AI-Powered Image Understanding," 2019.

[13] Unsloth AI, "Unsloth: 2x Faster LLM Fine-tuning," 2024.

[14] llama.cpp Contributors, "llama.cpp: Port of Facebook's LLaMA in C/C++," GitHub, 2024.

[15] Google, "TensorFlow Lite: On-Device Machine Learning," 2024.

[16] J. Hattie and H. Timperley, "The Power of Feedback," *Review of Educational Research*, vol. 77, no. 1, pp. 81–112, 2007.